

Do bad ideas spread exactly when fan-out times verification leak exceeds one?

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design written 2026-10-08; not yet agreed; no run started

1 Question

A persuasive but wrong artifact enters a group. Each agent that holds it teaches k others per generation (the fan-out of the routing rule), and each receiver’s verification lets a bad artifact through with probability a (the leak). If adoption decisions are independent across receivers, the number of new carriers per carrier per generation is ka , and a branching process dies out when $ka < 1$ and grows when $ka > 1$. We have measured a for three verification rules on reference-family bots (`findings/2026-10-06-sequential-verification.md`): 0.061 for the current rule (adopt if the paired mean over 200 games is positive), 0.003 for the sequential test, and 1 with no check. The question is whether the threshold $ka = 1$ is where a swarm’s safety actually turns, which would make “safe iff $ka < 1$ ” a design rule rather than a hope. A second question rides along: rejected prose still lowers learners by about 2.4 points (`findings/2026-10-07-rejected-prose-still-hurts.md`), a channel no verification touches. We should see harm without carriers, and that is the signature to look for.

2 Setting

One group of 8 agents, 20 generations, hosted model. Dials held fixed: topology one group; delivery deterministic; adoption replace-if-better; reaction rule read-everything; no migration; no credit; uniform budget. The two dials that vary are routing fan-out and verification. Sabotage: each delivered message is replaced, independently with probability $\varepsilon = 0.2$, by the 0-point bot with persuasive prose and the harness’s honest evidence (common random numbers across cells, so the sabotaged set is nested). This is the seeding rate; it is held fixed because the claim is about spread, not entry.

3 What we measure

Let C_t be the fraction of agents at generation t whose incumbent is the bad artifact or a descendant of it (the lineage graph records parents). The primary outcome is C_{20} and the shape of $t \mapsto C_t$; secondary is the group’s mean held-out self-play $\bar{Y}_t$, and the pair $(C_t, \bar{Y}_t)$ together is what separates the two channels: carriers falling to zero while $\bar{Y}_t$ still sits below the no-sabotage run is harm through prose. The quantity the design is built around is $R_0 = ka$ per cell, with a taken from the engine-only measurement (so the prediction is parameter-free: nothing is fitted on this run).

4 Prediction (2026-10-08)

verification	a	$k = 1$	$k = 7$
none	1	$R_0 = 1$: carriers persist, $C_{20} \approx 0.5$	$R_0 = 7$: everyone by generation
mean > 0 on 200 games	0.061	0.06: each seeding dies within 1–2 generations	0.43: dies within 3; $C_t < 0.1$
sequential test	0.003	0.003: no carriers beyond the seeded ones	0.02: same

And in every verified cell, $\bar{Y}_{20}$ is 1 to 3 points below the $\varepsilon = 0$ control although $C_{20} \approx 0$: the prose channel. What would make us drop the threshold claim: carriers persisting at $R_0 = 0.43$ (adoptions are not independent, e.g. a bad artifact that one agent adopts is then re-taught with new prose that passes more often), or no spread at $R_0 = 1$ without verification.

5 Design

Pilot: fan-out $k \in \{1, 7\} \times \text{verification} \in \{\text{none}, \text{mean} > 0\}$, plus an $\varepsilon = 0$ control at $k = 7$ with verification, 2 seeds, 10 generations: 10 runs. Full: $k \in \{1, 3, 7\} \times \text{verification} \in \{\text{none}, \text{mean} > 0, \text{sequential}\} \times 3$ seeds, 20 generations, paired on seeds and deals across cells, plus the control: 30 runs. Null: the stub in null mode with the same sabotage, where a for the verified rules is 0 by construction (the stub’s bad bot always loses), giving the carrier curve with no model. Figure: C_t against t , one panel per verification rule, lines by k , the predicted R_0 printed on each line; a second panel of $\bar{Y}_{20}$ against R_0 . Decision rule, fixed now: the threshold claim holds if, in every seed, cells with $R_0 < 0.5$ reach $C_t = 0$ within 5 generations of each seeding and cells with $R_0 > 2$ reach $C_t \geq 0.75$ by generation 10. The prose-channel claim holds if the verified cells’ $\bar{Y}_{20}$ is below the control’s with the paired interval excluding zero. Needs one small build: the sequential test as a verification policy (the analysis exists in `analysis/sprt.py`; the policy is thirty lines).

6 Cost

Pilot: 10 runs $\times$ 10 generations $\times$ (8 revise + up to 8 teach) $\approx$ 1,400 calls, a few dollars, under an hour in parallel. Full: 31 runs $\times$ 20 $\times$ 16 $\approx$ 10,000 calls, tens of dollars, one evening in parallel. Engine time: verification at 200 games per message dominates; the sequential arm is cheaper.

7 Nearest prior work

Byzantine-robust aggregation gives fraction thresholds for consensus under corrupted agents (Liu et al. 2026, arXiv 2604.17139; Lee et al. 2026, arXiv 2605.09076); memory-poisoning defenses report attack success rates (MAPLE-Guard, arXiv 2608.00426). A branching-process threshold for execution-verified adoption in a learning population, with the leak rate measured independently and the prose channel separated, was not found in the 2026-10-06 search.